

INTELLIGENT FREIGHT FORECASTING & VESSEL OPTIMIZATION SYSTEM

For Optimized Vessel Chartering & Bulk Cargo Procurement Across India's East Coast

Smart India Hackathon 2026 • Problem Statement SIH26006

FINAL CONSOLIDATED PRODUCT REQUIREMENTS DOCUMENT

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This document consolidates PRD v1.0 and Updated PRD v2.0. It preserves the original forecasting, data, API, dashboard, MLOps, execution, demo, risk, and SIH strategy details while incorporating the four expanded solution pillars: market-entry timing, vessel-type optimization, multi-voyage/idle-scenario management, and risk/sensitivity analysis.

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1. Executive Summary

India's East Coast bulk-cargo importers, charterers, freight brokers, and logistics planners face volatile freight markets and fragmented operational decisions. Teams often react to spot-market prices instead of systematically forecasting rates, coordinating vessel selection with port constraints, and planning multiple voyages together. The result is missed procurement windows, inefficient vessel choices, avoidable idle time, and procurement costs estimated at 15–25% above an optimal plan.

The proposed solution is an AI-driven Intelligent Freight Forecasting & Vessel Optimization System. It combines historical freight rates, fuel prices, commodity prices, AIS-derived vessel/queue signals, port constraints, origin-commodity availability, weather/seasonality, and machine-learning models to turn market data into an actionable procurement and chartering plan.

The system's central output is not merely a predicted freight rate. It answers: when should we enter the market, which vessel type should we use, which voyages should be bundled, how much could the plan save, and how does the recommendation change when market conditions shift?

Core Product Outputs

- 14-, 30-, and 90-day freight-rate forecasts with confidence intervals.
- Optimal market-entry recommendations such as Buy Now, Wait X Days, or Avoid Window.
- Ranked vessel-type recommendations: Handysize, Supramax, Panamax, Capesize.
- Port-feasibility filtering using draft, LOA, beam, cargo capacity, and congestion constraints.
- Multi-voyage contract/bundling plans for 3–6 week and 3–6 month planning horizons.
- Idle-time forecasts and cost/savings comparison against spot contracting.
- What-if sensitivity analysis for fuel, congestion, commodity prices, seasonality, and deadline changes.
- Explainable drivers using feature importance / SHAP and quantified uncertainty.

2. Problem Definition & Market Context

2.1 Operational Problems

- Daily market exploration without data-driven insight: procurement teams react to spot contracts and may miss optimal entry windows.
- Inefficient contract strategy: reliance on individual spot contracts rather than coordinated short-term (3–6 weeks) and medium-term (3–6 months) multi-voyage contracts.
- Geographic constraints: cargo originates from Australia, the US, Mozambique, Russia, and Indonesia, each with different transit times and infrastructure constraints.
- Vessel mismatch: no systematic selection between Handysize, Supramax, Panamax, and Capesize for a route/commodity/port combination.
- Freight volatility: weekly swings can reach roughly $\pm 20\%$; fuel shocks, congestion, weather, and seasonality amplify uncertainty.
- Procurement overruns: the PRDs identify a potential 15–25% premium versus an optimal decision.

2.2 Market Context

The original PRD frames India as handling roughly 600 million tonnes of cargo annually through ports, with East Coast ports representing about 35% of this volume. Freight costs are described as approximately 8–12% of imported commodity prices, making timing and chartering decisions economically significant.

Key stakeholders include bulk cargo importers (iron ore, coal, fertilizers, grains), ship charterers and freight brokers, port authorities, logistics companies, and supply-chain/procurement teams.

3. Product Vision, Goals & Success Metrics

Vision: Enable data-driven freight procurement and vessel-chartering decisions on India's East Coast, reducing logistics cost and idle time while making recommendations transparent enough for operational teams to trust.

Goal	Target / Acceptance Signal
Forecast quality	MAPE <15% on validated test/backtest data; directional accuracy also reported.
Forecast horizons	14, 30, and 90 days.
Procurement impact	Demonstrate quantified savings against a spot-market baseline; target around 10–15% in the PRD.
Vessel optimization	Rank feasible vessel classes with cost + port-fit rationale.
Idle optimization	Demonstrate measurable idle-time reduction in a multi-voyage scenario.
API performance	<500 ms p95 for a single forecast; 100+ concurrent requests target.
Reliability	99.5% target service uptime for a production-style deployment.
Explainability	Top drivers, uncertainty, and scenario effects visible to users.

4. Users & Core Use Cases

User	Primary Need	System Output
Bulk-cargo procurement team	Choose when/how much to procure.	Forecast + optimal entry window + expected savings.
Ship charterer / broker	Choose vessel and contract strategy.	Vessel ranking + route cost + multi-voyage plan.
Logistics planner	Coordinate origins, destinations, and timing.	Voyage schedule + idle forecast + port feasibility.
Supply-chain manager	Assess risk before committing.	Confidence intervals + sensitivity analysis + anomalies.
Judge / evaluator	Verify technical feasibility and impact.	Backtests, explainability, architecture, live demo.

Core Use Cases

- Forecast freight for a selected origin-destination route, commodity, volume, and horizon.
- Identify the best procurement window from the forecast and confidence interval.
- Select the most suitable vessel class for volume, port infrastructure, and schedule.
- Build a coordinated multi-voyage contract plan and compare it with spot contracting.
- Test market shocks such as fuel +5%/+10%, congestion surge, monsoon disruption, or deadline changes.
- Inspect historical rates, forecast drivers, model components, and backtest performance.

5. Solution Architecture — Four Pillars

5.1 Pillar A — Optimal Market Entry Timing

The forecasting layer blends Prophet, LSTM, and XGBoost. The original ensemble is 0.25 Prophet + 0.40 LSTM + 0.35 XGBoost, with weights tuned through time-series cross-validation and potentially adjusted per route/commodity.

- Prophet: trend, seasonality, holidays, interpretable baseline.
- LSTM: nonlinear temporal dependencies using a 30-day lookback.
- XGBoost: lagged effects, thresholds, feature interactions, and interpretable importance.
- Output: predicted freight rate, 95% uncertainty interval, directional movement, and recommended action.

5.2 Pillar B — Vessel Type Optimization

An XGBoost multi-class classifier/ranking model evaluates Handysize, Supramax, Panamax, and Capesize candidates using route, commodity, volume, port compatibility, draft margin, transit time, freight, port fees, insurance, utilization, and schedule compliance.

- Infeasible vessels are filtered before ranking; e.g., a vessel exceeding a selected port's draft/LOA constraints should not be recommended.
- Output includes ranked vessel types and a cost breakdown rather than a bare classification.

5.3 Pillar C — Idle Scenario & Multi-Voyage Optimization

The optimizer shifts the decision from isolated spot charters to coordinated multi-voyage planning. A dynamic-programming prototype is the primary feasibility path; reinforcement learning may be explored only if time and data permit.

- Inputs: procurement plan, vessel inventory/availability, voyage requirements, timing constraints.
- Outputs: bundled voyage sequence, contract windows, estimated idle periods, and cost/savings versus spot.
- Illustrative PRD scenario: coordinate Mozambique (Weeks 1–6), Russia (Weeks 7–12), Australia (Weeks 13–18), reducing idle time from 35% to 8% and illustrating \$125K savings on a \$725K plan.

5.4 Pillar D — Risk Mitigation & Sensitivity Analysis

- Forecast uncertainty: Prophet confidence intervals; LSTM MC-dropout uncertainty; XGBoost explanations/SHAP-based diagnostics.
- Anomaly detection for fuel spikes, congestion surges, and seasonal regime shifts.
- Interactive what-if controls: fuel price, congestion, commodity price, monsoon/seasonality, deadline.
- Recommendations update when scenario variables change.

6. Geographic, Port & Vessel Domain Model

6.1 East Coast Port Constraints

Port	Draft (m)	Max LOA (m)	Beam (m)	Cargo Cap	Main	Suitable Vessel
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					Commodities	Types
Paradip	13.0	240	32.2	140K+ t	Iron ore, coal	Capesize, Panamax
Vizag	12.5	238	32.2	120K t	Iron ore, manganese	Panamax, Supramax
Gangavaram	12.0	200	32.0	80K t	Coal, lignite	Supramax, Handysize
Gopalpur	10.5	180	30.0	60K t	Coal, pellets	Handysize, Supramax
Dhamra	11.0	210	32.0	100K t	Iron ore, coal	Panamax, Supramax
Sagar-Sandheads	9.5	170	28.0	50K t	General cargo	Handysize, Small Panamax
Haldia	8.5	160	27.5	40K t	Coal, containers	Handysize

6.2 Origin–Commodity Flows

Origin	Indicative Commodity Mix	Target East Coast Ports	Typical Vessel / Transit
Australia	Iron ore 70–75%; coal 15–20%	Paradip, Vizag	Capesize; ~14–16 days
US East Coast	Coal ~90%; grains ~10%	Mixed	Supramax/Panamax; ~28–32 days
Mozambique	Coal, mineral sands	Gopalpur, Gangavaram	Supramax/Handysize; ~10–12 days
Russia	Coal, fertilizers	Dhamra, Paradip	Mixed; ~12–16 days
Indonesia	Coal, nickel ore	Gopalpur, Gangavaram, Vizag	Handysize/Supramax; ~7–9 days

7. Data Sources & Data Pipeline

Data	Source / Approach	Refresh	Use
Freight rates / indices	BDI, SCFI, maritime/Kaggle datasets; route-level data where available	Daily	Target variable + benchmark.
Fuel / bunker prices	IEA/EIA, Bloomberg/market APIs where available	Daily	Cost driver + volatility.
AIS vessel data	MarineTraffic, Kaggle, OpenStreetMap-derived context	Real-time / ~6-hourly where available	Queue length, vessel distribution, supply proxy.
Port congestion	Port authority APIs/dashboards + AIS proxies	Daily	Wait time, queue, congestion score.
Commodity prices	Trading Economics, CEIC, commodity datasets	Daily	Demand/price driver.
Weather / sea state	NOAA, Met India, Copernicus	~3-hourly	Disruption / seasonal features.
Port constraints	Port authority websites/PDFs + DGFT shipping indices	Quarterly / curated	Draft, LOA, beam, capacity.
Origin availability	Mining/company reports, USGS, SIMA indices, Russian port data	Weekly	Supply forecast by origin.
Seasonal indicators	Feature engineering: holidays, monsoon, month, day-of-week	Static / derived	Seasonality and regime effects.

7.1 Pipeline

- Ingestion: Airflow DAGs fetch daily/hourly data; Kafka is reserved for real-time streams where justified.
- Validation: Great Expectations checks schema, ranges, missingness, duplicates, and freshness.
- Storage: PostgreSQL + TimescaleDB for structured/time-series data; Redis for cache.
- Preprocessing: Pandas/Polars; missing-value handling, outlier treatment, normalization, lag construction.
- Feature management: Kedro pipeline and DVC for reproducible datasets/features.
- Training: weekly retraining; MLflow tracks experiments, metrics, and model versions.

8. Machine Learning & Optimization Architecture

8.1 Forecast Models

Model	Role	Key Configuration / Features	Strength	Risk
Prophet	Baseline time-series decomposition	Yearly seasonality; changepoint_range 0.9; prior scale 10	Fast, interpretable	Limited exogenous modeling unless regressors are added.
LSTM	Temporal/nonlinear forecasting	2 layers: 64→32; Dense 16→1; 30-day lookback; dropout 0.2; Adam; MSE; early stopping	Captures nonlinear temporal patterns	Data hungry / less interpretable.
XGBoost	Feature-driven forecast	Lagged rate, fuel, congestion, commodity, month, volatility, queue, seasonal flag	Strong tabular interactions + feature importance	Can overfit sparse route/commodity combinations.

8.2 Evaluation & Backtesting

- Primary metric: MAPE; target <15% (equivalent to >85% accuracy framing used in the PRDs).
- Secondary metrics: RMSE, directional accuracy, and prediction-interval coverage.
- Validation: chronological train/validation/test split (70/15/15) and walk-forward backtesting.
- Compare against simple moving average / spot baseline to prove incremental value.
- Report model performance per horizon and route/commodity where data volume permits.

8.3 Ensemble

Final forecast = $0.25 \times \text{Prophet} + 0.40 \times \text{LSTM} + 0.35 \times \text{XGBoost}$. The weights are the initial baseline from the PRDs and should be validated rather than presented as inherently optimal.

8.4 Optimization Modules

- Vessel classifier: XGBoost multi-class ranking based on cost, port fit, transit, utilization, and schedule.
- Idle optimizer: dynamic programming as MVP; RL only as an optional experiment.
- Sensitivity engine: recompute forecast/recommendation under scenario perturbations.
- Recommendation engine: converts forecast + risk + operational constraints into Buy/Wait/Avoid and savings estimates.

9. Functional Requirements

ID	Requirement	Acceptance
FR-01	Route & cargo selection	User selects origin, destination, commodity, and volume; auto-suggest supports route search.
FR-02	Freight forecast	Generate 14/30/90-day forecasts with 95% confidence bands.
FR-03	Market-entry recommendation	Show Buy Now / Wait X Days / Avoid Window with rationale and expected savings.
FR-04	Model transparency	Allow historical + Prophet/LSTM/XGBoost + ensemble views.
FR-05	Feature importance	Show top drivers such as fuel, congestion, commodity price, seasonality.
FR-06	Vessel recommendation	Rank feasible vessel classes and expose cost/port-fit breakdown.
FR-07	Port constraints	Filter vessels that violate draft, LOA, beam, or cargo-capacity limits.
FR-08	Multi-voyage optimization	Generate coordinated voyage/contract plan and idle estimate.
FR-09	Sensitivity analysis	Update forecast and recommendation

		under user-defined scenarios.
FR-10	Historical analysis	Display historical route rates for selected dates/granularity.
FR-11	Origin availability	Expose commodity availability forecasts by origin where data exists.
FR-12	Operational fallback	Allow a local/canned dataset and precomputed predictions for offline demo.

10. API Specification

Method	Endpoint	Purpose
POST	/api/v1/forecast	Forecast route/commodity/volume for selected horizon; return rate, CI, recommendation, factors.
GET	/api/v1/historical-rates	Historical time series by route, commodity, date range, granularity.
GET	/api/v1/routes	Supported origin-destination routes and commodities.
POST	/api/v1/procurement-recommendation	Optimize procurement timing under budget, commodities, timeline, and risk tolerance.
POST	/api/v1/vessel-recommendation	Rank vessel types with cost breakdown and port fit.
POST	/api/v1/multi-voyage-optimization	Bundle procurement/voyages; return schedule + idle forecast.
GET	/api/v1/port-constraints?port=paradip	Return draft, LOA, beam, congestion score, queue length.
POST	/api/v1/sensitivity-analysis	Live what-if recalculation.
GET	/api/v1/origins/available	Commodity supply/availability forecast by origin.

10.1 Example Forecast Contract

Request: {"route":"Paradip-Rotterdam","commodity":"iron_ore","volume_tonnes":50000,"forecast_horizon_days":30}

Response: {"predicted_rate_usd_per_tonne":28.50,"confidence_interval":[27.2,29.8],"recommendation":"Optimal time to procure","factors":{"fuel_price_impact":0.45,"seasonality":0.30,"congestion":0.15,"commodity_price":0.10}}

10.2 API Non-Functional Requirements

- p95 forecast latency <500 ms for a single request.
- 100+ concurrent forecast requests as a load-test target.
- Meaningful validation and error responses; no unhandled 500 errors.
- Structured logging of forecast inputs, outputs, timestamps, model version, and scenario.
- Swagger/OpenAPI documentation generated by FastAPI.

11. Frontend / Dashboard Requirements

- Route selector with origin, destination, commodity, and volume.
- 90-day interactive forecast chart with historical data, confidence bands, and model toggles.
- Recommendation panel showing today's optimal action, rationale, expected savings, and risk.
- Feature-importance / driver chart; SHAP view where feasible.
- Vessel ranking table with feasibility badges and cost breakdown.
- Port-constraint panel showing draft, LOA, beam, queue, congestion score.
- Idle-risk heat map and multi-voyage Gantt chart with idle periods flagged.
- Sensitivity sliders for fuel, congestion, commodity price, seasonality/monsoon, and deadline.

- Procurement optimizer form: budget, commodities, timeline, risk tolerance.
- Responsive TailwindCSS layout for desktop/tablet/mobile.

12. Technology Stack & Architecture

Layer	Technology	Purpose
Frontend	React 18, TailwindCSS, Recharts, Axios	Dashboard, charts, interactions.
Backend	FastAPI, Uvicorn, Pydantic	Async API, validation, inference.
Forecasting	Prophet, TensorFlow/PyTorch, XGBoost, scikit-learn	Forecast + ranking models.
Optimization	Python DP; optional RL	Multi-voyage bundling.
Data	PostgreSQL + TimescaleDB	Transactional + time-series storage.
Cache	Redis	Prediction/query caching.
Ingestion	Apache Airflow, Python requests/aiohttp; Kafka optional	Scheduled and streaming ingestion.
Features	Kedro, Pandas/Polars, feature-engine	Reproducible transformations.
MLOps	MLflow, DVC	Experiments, model/data versioning.
Monitoring	Prometheus, Grafana; ELK where needed	Metrics and logs.
Testing	Pytest, Jest, Great Expectations, Locust	Unit, integration, data and load tests.
Quality	Black, Ruff, pre-commit	Formatting and linting.
Deployment	Docker, Compose, GitHub Actions; Render/Railway MVP; AWS production	Reproducible deployment and CI/CD.

MVP principle: use the smallest reliable subset that can be demonstrated. Kafka, Kubernetes, ELK, full RL, and proprietary real-time feeds are optional production extensions—not mandatory hackathon blockers.

13. Data Model & Repository Structure

13.1 Core Tables

- `freight_rates`: route, commodity, date, rate, source, granularity.
- `fuel_prices`: date, benchmark, price, volatility.
- `port_specs`: port, draft, max_loa, beam, cargo_capacity, commodity support.
- `vessel_specs`: vessel class, draft, LOA, beam, capacity, typical cost.
- `port_observations`: queue length, berth availability, wait time, congestion score.
- `origin_commodities`: origin, commodity, availability, historical volume, forecast.
- `forecast_runs`: model version, inputs, predictions, intervals, recommendation.
- `voyage_plans`: procurement requirements, vessel, origin/destination, timing, idle estimate, cost.

13.2 Recommended Repository

sih26006-freight-forecasting/

```

├── data/           # Raw/processed CSV, Parquet
├── notebooks/      # EDA + model experiments
├── src/
│   ├── features/   # Feature engineering / Kedro
│   ├── models/     # Prophet, LSTM, XGBoost
│   ├── optimization/ # Vessel + multi-voyage optimization
│   ├── api/        # FastAPI
│   └── utils/
├── frontend/       # React dashboard
├── airflow/        # DAGs
├── tests/
└── docker-compose.yml

```


14. Pre-Hackathon Roadmap — 6 Weeks

Week	Goal	Key Work	Deliverables
1–2	Data foundation	Collect 2+ years freight data; fuel/commodity/AIS; curate 7-port constraint matrix; origin mapping; EDA.	Clean dataset, EDA notebook, feature list, port/origin matrices.
3–4	Models + optimization	Prophet/LSTM/XGBoost; time-series CV; vessel classifier; DP optimizer prototype; backtest.	Tracked models, validation report, vessel ranking, multi-voyage proof-of-concept.
5	Backend + database	FastAPI endpoints; PostgreSQL/TimescaleDB; Airflow refresh; inference service; load test.	Working API, schema, DAG, Swagger, load-test result.
6	Frontend + deployment	Dashboard, vessel table, Gantt, sensitivity UI, Docker, CI/CD, deployment.	Live MVP, Docker stack, documentation, demo-ready dataset.

Week 1–2 Detailed Data Tasks

- Download/assemble BDI/SCFI and route-relevant maritime data.
- Collect historical fuel and commodity prices.
- Compute AIS queue lengths and vessel-type distributions where data is available.
- Curate port draft/LOA/beam/capacity constraints for all seven East Coast ports.
- Build Australia/US/Mozambique/Russia/Indonesia origin-commodity mapping.
- Engineer day-of-week, month, holiday, monsoon, lag, volatility, congestion, and queue features.

Week 3–4 Detailed Model Tasks

- Prophet baseline and seasonal/changepoint tuning.
- LSTM training with normalized inputs and early stopping.
- XGBoost with lagged rates, fuel volatility, queue/congestion, commodity, and seasonal features.
- Walk-forward validation; no look-ahead leakage.
- Ensemble weighting and holdout validation.
- Vessel classifier and feasibility rules.
- DP multi-voyage optimizer; quantify idle and cost improvement.

15. 72-Hour Hackathon Execution Plan

Time	Focus	Must Finish
Hour 0–12	Integration & validation	Forecast endpoint, DB, routes, dashboard smoke test, model loading, CORS, fallback data.
Hour 12–36	Core product	Recommendation engine, vessel ranking, port filtering, feature importance.
Hour 36–48	Optimization + scenarios	Multi-voyage demo, idle metric, sensitivity sliders.
Hour 48–60	Testing & docs	15+ core tests, integration tests, README, architecture, canned demo.
Hour 60–72	Presentation	Dry run, pitch deck, impact metrics, Q&A, offline backup.

Priority rule: a stable four-pillar MVP beats an overextended system with half-finished features. If time collapses, preserve forecasting + vessel feasibility/ranking + one deterministic multi-voyage scenario + sensitivity demo, and make the rest explicit future work.

16. Testing, Validation & Quality Gates

Area	Minimum Gate
Data	Schema/range/freshness checks; no leakage; documented missing-data policy.
ML	MAPE <15% target on test set; RMSE, direction accuracy, CI coverage reported.
Backtest	Walk-forward evaluation and comparison with simple baseline.
API	All core endpoints return validated responses; <500 ms p95 forecast target.
Load	100 concurrent requests target using Locust.
Frontend	Route selection, charts, ranking, Gantt, sensitivity all smoke-tested.
Code	>80% test coverage target; Ruff/Black/pre-commit clean.
Deployment	docker-compose up works locally; public deployment plus localhost/offline fallback.

17. Risks, Fallbacks & Demo Reliability

Risk	Mitigation / Fallback
Overfitting / leakage	Chronological splits, walk-forward validation, regularization, baseline comparison.
Insufficient data	Use public datasets and clearly label proxy features; reduce route/commodity scope.
Real-time API unavailable	Use cached/AIS-derived proxy metrics and preloaded observations.
LSTM too slow	Train in Colab/Kaggle; fall back to Prophet + XGBoost for demo if necessary.
Port data unavailable	Manually curate constraint matrix from public port information; use AIS queue proxy.
RL too complex	Use deterministic dynamic programming for the hackathon MVP.
Dashboard/internet failure	Local DB, canned forecast JSON, localhost demo, recorded backup.
API latency >500 ms	Redis caching, precomputed forecasts, profiling, model serialization, horizontal scaling if needed.
Model explanation weak	Use XGBoost feature importance/SHAP; expose uncertainty rather than claiming causality.

Important credibility rule: all numeric impact examples (accuracy, savings, idle reduction, market value) must be clearly labeled as measured backtest results, illustrative scenarios, or targets. Do not present an illustrative number as a measured result.

18. SIH Winning Strategy & Differentiation

18.1 Why the Product Is Different

- Forecasting + optimization: it does not stop at predicting freight; it converts predictions into procurement and chartering actions.
- Four-pillar workflow: timing, vessel choice, multi-voyage coordination, and risk/sensitivity are connected in one decision system.
- Domain specificity: East Coast ports, bulk commodities, origin flows, port constraints, and vessel classes are explicitly modeled.

- Explainable AI: feature drivers and uncertainty help users understand why the system recommends Buy/Wait/Avoid.
- Operational feasibility: vessel recommendations respect real port constraints rather than optimizing cost in isolation.
- Realistic data strategy: the MVP can work with public data and transparent proxies rather than requiring proprietary broker data.

18.2 Judge-Facing Metrics

- MAPE <15% on a documented test/backtest set.
- Quantified savings against spot/simple baseline.
- Idle reduction in a reproducible multi-voyage scenario.
- Port feasibility filtering shown live.
- Sensitivity response shown live for a fuel/congestion shock.
- API latency and load-test evidence.
- Clean code, tests, reproducible Docker setup, and offline demo fallback.

19. Demo Script & Judge Q&A

19.1 15-Minute Demo

Segment	Time	Demo
Problem framing	2 min	Show freight volatility, reactive procurement, vessel/port mismatch, and idle problem.
Forecast	4 min	Select route/commodity; show 90-day forecast, confidence band, model components and drivers.
Vessel optimization	2 min	Show ranked vessel types and port-constraint filtering.
Multi-voyage plan	3 min	Show 3-voyage Gantt, idle reduction, and savings versus spot.
Sensitivity	2 min	Change fuel +10% or congestion; show updated forecast/recommendation.
Impact + Q&A	2 min	Show measured backtest metrics, scalability, fallback, and roadmap.

19.2 Anticipated Judge Questions

Q: What if fuel prices spike 30% overnight?

A: Use volatility/anomaly features and scenario sensitivity; confidence intervals widen when uncertainty rises. Retraining and refreshed features allow adaptation. Do not claim exact spike behavior unless validated.

Q: How do you get proprietary congestion data?

A: The MVP uses AIS-derived queue length, public port wait/berth information, and other transparent proxies. The model is designed to accept better data sources later.

Q: Why not just use a moving average or broker?

A: Backtesting should demonstrate whether the ensemble improves error and decision value. The system additionally exposes uncertainty, drivers, and operational constraints.

Q: How does this scale?

A: Route/commodity modules can be trained independently; Airflow can parallelize retraining; FastAPI is stateless; Redis caches repeated queries; database indexes support time-series retrieval.

Q: Why four vessel types?

A: They represent the main bulk-vessel classes in the PRD and make the recommendation concrete while allowing port constraints to eliminate infeasible choices.

Q: Why dynamic programming instead of RL?

A: DP is more deterministic, explainable, and feasible for a hackathon. RL is an optional extension once a richer simulator and sufficient data exist.

20. Resources, Dependencies & Hardware

20.1 Reference Data / Services

- Kaggle maritime freight datasets for BDI/SCFI and related historical shipping data.
- EIA/IEA historical fuel-price datasets or equivalent accessible public market data.
- MarineTraffic AIS services where access is available; otherwise public/Kaggle AIS subsets.
- Trading Economics / CEIC / commodity datasets where licensing/access permits.
- Port authority websites/PDFs and DGFT shipping indices for port constraints.
- NOAA / India Meteorological / Copernicus weather sources where required.

20.2 Development & Infrastructure

- Local: VS Code, Git, Docker Desktop.
- MVP hosting: Render/Railway; production architecture can move to AWS EC2 + RDS.
- MLOps: MLflow; DVC; optional W&B academic/free tier.
- Monitoring: Grafana Cloud/free tier and local Prometheus.
- Training: approximately 4-core CPU and 8 GB RAM baseline; GPU/Colab/Kaggle can accelerate LSTM.
- Inference: approximately 2-core CPU and 2 GB RAM for MVP; model caching can reduce latency.
- Storage: approximately 5 GB for initial model/data artifacts, increasing with AIS history.

21. Final Hackathon Acceptance Criteria

- ✓ Live forecast demonstrated on at least 3 routes/origins without crashes.
- ✓ Ensemble achieves MAPE <15% on a documented test/backtest dataset, or the actual measured result is transparently reported if the target is not achieved.
- ✓ Procurement recommendation is actionable and backed by forecast + uncertainty + drivers.
- ✓ Vessel recommendation ranks feasible vessel types and rejects incompatible port/vessel combinations.
- ✓ Multi-voyage demo produces a coordinated schedule and quantifies idle/cost change against a defined baseline.
- ✓ Sensitivity analysis visibly changes forecast/recommendation for a scenario such as fuel +10%.
- ✓ API p95 latency target <500 ms is tested and reported.
- ✓ Test coverage target >80% for production code, with unit + integration coverage of critical paths.
- ✓ docker-compose setup works locally and an offline/canned demo is available.
- ✓ README contains architecture, setup, data assumptions, API docs, model metrics, and limitations.
- ✓ Every headline savings/accuracy metric is backed by a reproducible calculation or explicitly labeled as a target/illustrative scenario.

Appendix A — Quick Reference

Metric / Component	Final Target / Baseline
Forecast horizons	14 / 30 / 90 days
Forecast quality target	MAPE <15%
Ensemble	0.25 Prophet + 0.40 LSTM + 0.35 XGBoost
Vessel classes	Handysize / Supramax / Panamax / Capesize
East Coast ports	7 ports in MVP constraint matrix
Origin focus	Australia / US / Mozambique / Russia / Indonesia
Commodities	Iron ore / coal / fertilizers / grains + mapped flows
Forecast API	<500 ms p95 target

Concurrency	100+ concurrent requests target
Retraining	Weekly; data refresh daily where applicable
MVP deployment	Docker + Render/Railway; localhost fallback
Production direction	AWS + scalable stateless FastAPI + managed DB/cache

Appendix B — Version Consolidation Note

Version 3.0 is the consolidated SIH 2026 PRD. PRD v1.0 supplied the detailed baseline forecasting architecture, full API/dashboard specification, technology stack, six-week execution plan, 72-hour plan, risk controls, demo script, judge Q&A, and winning strategy. Updated PRD v2.0 supplied the expanded operational scope: four solution pillars, seven-port constraint matrix, origin flows, vessel classifier, multi-voyage/idle optimization, sensitivity analysis, expanded endpoints, and updated acceptance criteria. This final version combines those elements into a single implementation-oriented specification.